

Mathematics

A Cultural Heritage

Lecture 4

New Kinds of Math – Part 1: Infinity

Infinity

- **Infinity:** Not a number
- Ancient Greeks: Accepted potential infinity
 - **Potential Infinity:** $\{1, 2, 3, \dots, N\}$ where N can be any large number
 - **Actual Infinity:** All whole numbers
- Paradoxes of Zeno (c. 490BC-430BC):
 - **Achilles and the tortoise**
 - **Dichotomy paradox:** “That which is in locomotion must arrive at the half-way stage before it arrives at the goal.”...(conclusion: any motion is impossible!)

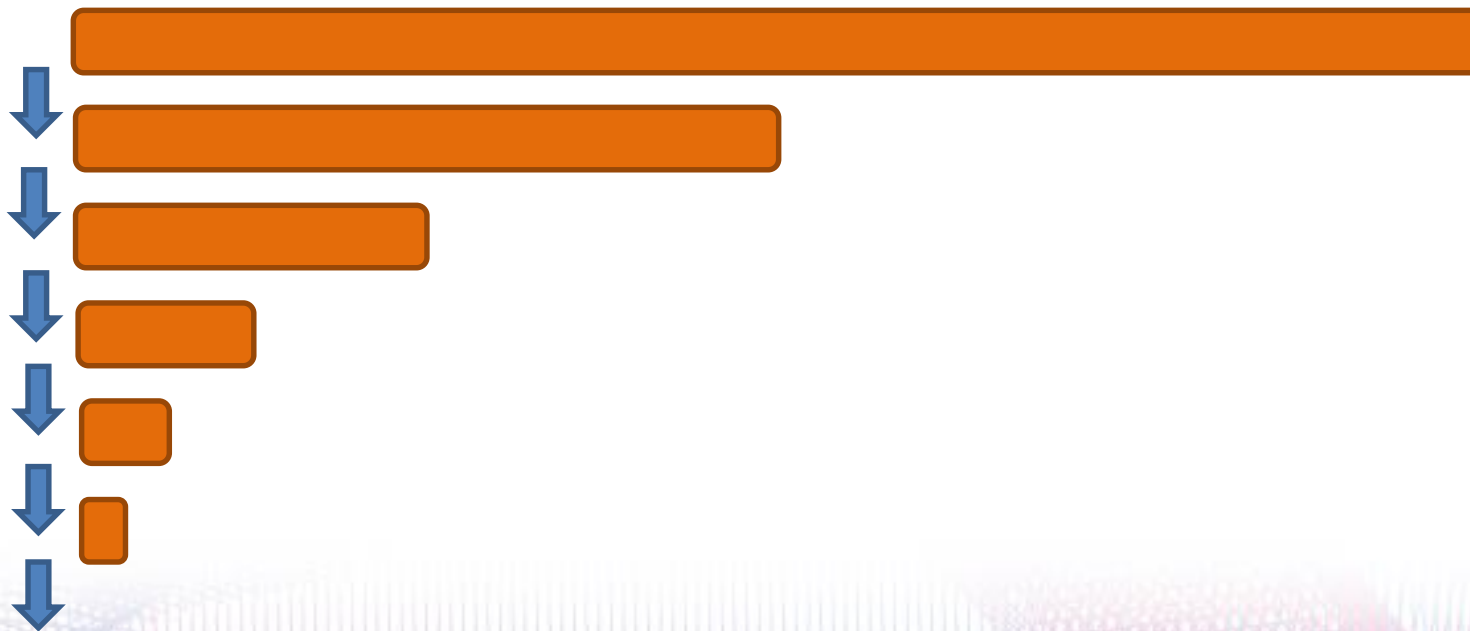
Infinity in Chinese History

- The following statement is attributed to the Chinese debater and philosopher 惠施 (**Hui Shi**, c.370BC-310BC)

「一尺之棰，日取其半，萬世不竭」

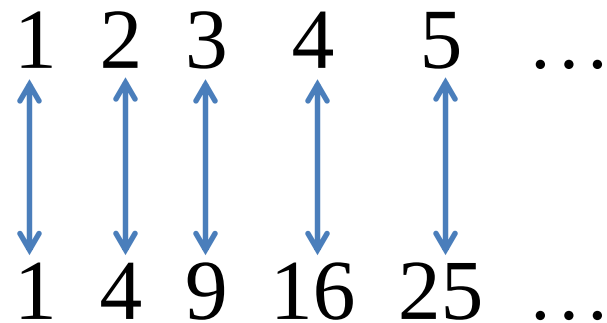
[Translation: “If from a foot-long stick you take half of it every day, in a myriad ages it will not be exhausted.”]

— from *Tianxia*, Zhuangzi (《莊子·天下篇》)



Oddities of Infinity

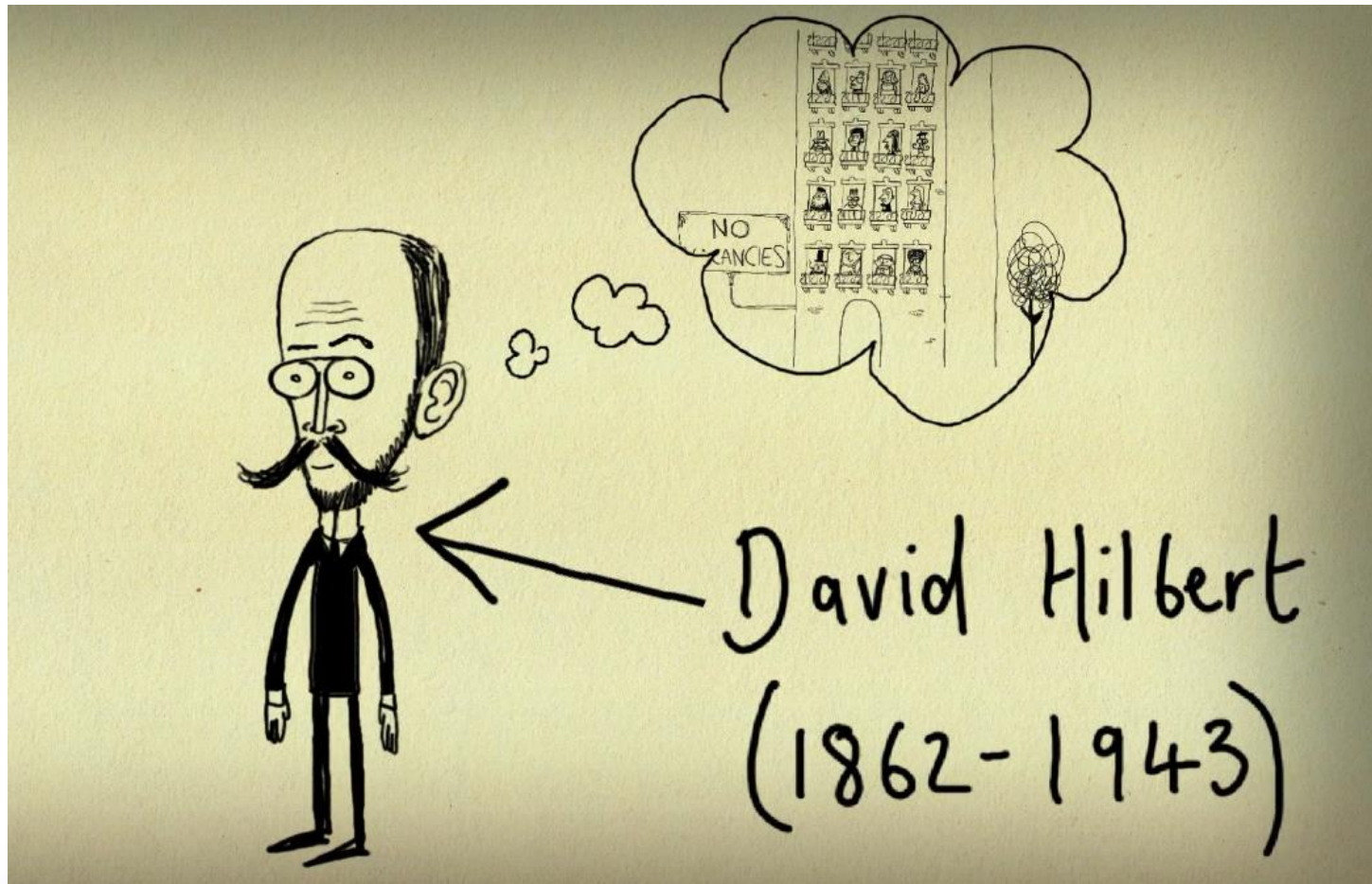
- Galileo Galilei (1564–1642): As many perfect squares as whole numbers:



- But more whole numbers than perfect squares?

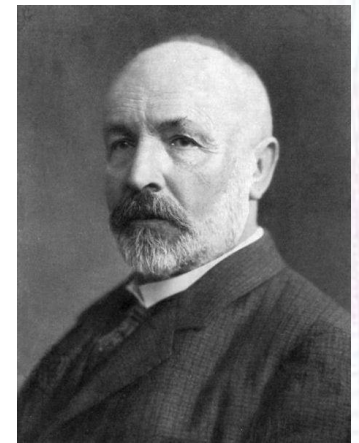
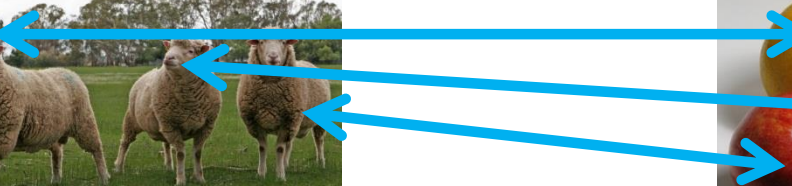
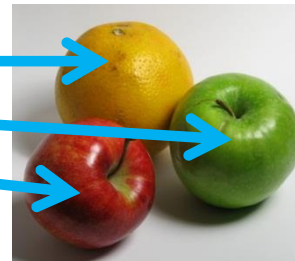
More Strange behavior

- Video: Hilbert's Infinite Hotel



Understanding Infinity

- **Georg Cantor** (1845 –1918): Invented set theory
- **Set**: Collection of objects; objects called **elements**.
- Empty set: No elements
- Finite set: Correspondence to a whole number.



Georg Cantor (1845-1918)
[source: wikipedia]

Understanding Infinity

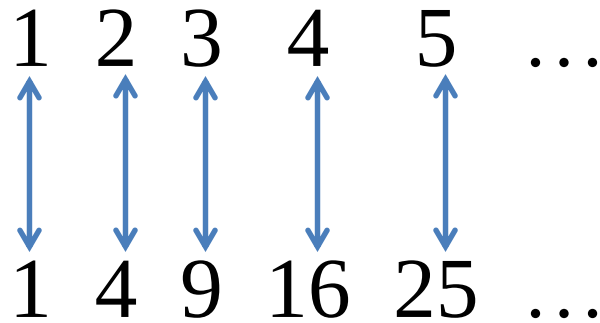
- Call A **subset** of B if B contains all elements of A .
- **Infinite sets**: Subset in 1-1 correspondence to whole numbers
- Alternatively: Smaller subset in 1-1 correspondence
- We say two sets have same **cardinality** if 1-1 correspondence between them.

Acceptance of Infinity

- Cantor's theory **rejected** in his time
- Church thought work **challenged uniqueness** of infinity of God.
- Only later accepted.
- **David Hilbert** (1862–1943): Defended Cantor's work: *“No one shall expel us from the Paradise that Cantor has created.”*

Potential vs. Actual Infinity

- Galileo Galilei (1564–1642):



- But “zero percent are squares”
- Consider number of squares up to N :

$$\frac{\# \text{ squares} \leq N}{N} \approx \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}$$